1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In tumors with RASSF1A loss, increased genomic instability and neoantigen burden recruit tumor-infiltrating lymphocytes but concurrently upregulate LAG3-mediated signaling, driving T cell exhaustion and compromising anti-tumor immunity; consequently, high TIL density will not translate into improved overall survival in RASSF1A-deficient tumors with elevated LAG3, when compared with RASSF1A-wildtype tumors, suggesting that these tumors may benefit from LAG3-targeted checkpoint blockade to restore TIL effectiveness.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	RASSF1A, LAG3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

RASSF1A: RASSF1A (a tumor suppressor frequently silenced by promoter methylation) to immune context via LAG3: RASSF1A loss plausibly drives tumor programs (genomic instability, altered signaling/metabolism) that promote T cell dysfunction or checkpoint upregulation, with LAG3 as a candidate mediator.

LAG3: LAG3: RASSF1A loss plausibly drives tumor programs (genomic instability, altered signaling/metabolism) that promote T cell dysfunction or checkpoint upregulation, with LAG3 as a candidate mediator.

3. Clinically Testable Predictions

Prediction 1: In silico TCGA analyses: stratify tumors by RASSF1A status (expression/methylation), TIL density, and LAG3; assess OS and interaction effects;

Prediction 2: In vitro: manipulate RASSF1A in breast cancer cell lines; evaluate genomic instability markers, neoantigen-related signaling, and co-culture with T cells to monitor activation/exhaustion markers (including LAG3 axis);

Prediction 3: In vivo: use RASSF1A-knockout murine models to measure TIL infiltration and LAG3 expression; test anti-LAG3 therapy for TIL reinvigoration and tumor control;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions